

Chapter 1

Basic Definitions

Definition 1.1 (Support of a function). $\forall f \in C(\mathbb{R})$, let

$$\text{supp}(f) = \overline{\{x \in \mathbb{R} : f(x) \neq 0\}}$$

Definition 1.2 (Compactly Supported Continuous Functions).

$$C_0(X) := \{f \in C(X) : \text{supp}(f) \text{ is compact}\}$$

Chapter 2

Main Theorem

Theorem 2.1 (Weierstrass Theorem). $\forall f \in C([a, b])$, $\forall \varepsilon > 0$ there exists a p , polynomial, such that $\|f - p\|_\infty < \varepsilon$.

Proof: Let $f(x) = f(\frac{1}{b-a}(x-a))$, f is now a function from $[0, 1] \rightarrow \mathbb{R}$. Let

$$g(x) = f(x) - f(0) + (f(0) - f(1))x.$$

This change in $g(0) = g(1) = 0$

□

Chapter 2

References